Verbalizable Causal Channels Are Model- and Task-Dependent in Open Language Models

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Abstract

A 2026 interpretability result proposes that the verbalizable subspace read out by a fitted Jacobian lens forms a global workspace *in language models: sparse, broadcast, and causally privileged, such that deleting its content selectively destroys multi-step behavior while sparing direct recall. This paper reports a three-phase, preregistered replication program on open-weights models (OLMo 3.1 32B Think and Instruct; Qwen 3.6 27B) whose central technical finding is about the* instrument*: the paper's output-protection contract protects output* labels*, not output* geometry*, and 28–42% of the activation energy removed by the "protected" ablation lies inside the protected output span. Under a span-safe arm that removes this leak by construction, a J-specific heavy tail of per-item damage survives at confirmatory tier on Qwen (tail-rate excess* $+0.096$ *over an exactly rank-and-energy-matched control,* $p \approx 10^{-5}$*; independent replication* $+0.102$*) at roughly one third of its label-protected size. A preregistered bridge-protection test confirms mediation on Qwen: protecting the true intermediate ("bridge") entity's directions rescues composed-item damage by* $+0.43$ *nats relative to a matched counterfactual bridge, and substituting the counterfactual bridge's content at matched dose is catastrophically worse than deletion (*-4.05 *nats) — the model demonstrably reads what sits in the channel. OLMo shows none of this: its damage is organized by answer accessibility rather than task shape, and its rescue contrast is mildly negative. The workspace-word test fails on both clusters — the composed-vs-direct contrast does not appear on in-context task banks — so the supportable noun is* knowledge-access channel*, not workspace. All primary quantities reproduce clean-room from frozen raw data to* $\leq 5 \times 10^{-4}$*, including one full GPU cell bit-exactly.*

I. The Problem

The claim under test comes from the July-2026 paper *"Verbalizable Representations Form a Global Workspace in Language Models"* [Anthropic, 2026]: a small, verbalizable subspace of the residual stream — read out by a fitted Jacobian lens [Anthropic, 2026] — is *causally privileged*, such that deleting its content selectively destroys composed (multi-hop) behavior while sparing direct recall. The claim matters because it is the strongest mechanistic-interpretability result to date connecting a *readable* representation to a *causally load-bearing* one, and because "workspace" imports a functional architecture — a shared bus between specialist processes — that circuits-level results do not usually claim.

[TODO: 1–2 paragraphs situating the replication program: Phase 1 exploratory lab (superseded by errata), Phase 2 preregistered confirmatory campaign on the audited assay, Phase 3 mechanism + generalization. Cite the governing preregistrations.]

Separating "the model lost the fact" from "the model lost the ability to say anything" from "the model was injured proportionally to how much of anything was removed" is the entire technical problem; §II shows that each confusion occurred in practice and was caught by an instrument audit.

II. Assay Repair

[TODO: Port the assay-repair story: each repair exists because an instrument fault produced, or would have produced, a false claim.]

- **Output protection done properly.** Per-position exclusion of the clean top-k token rows from J-selection; the open question it leaves — label exclusion bounds *selection*, not *geometry* — becomes §IV.

- **The exact instantaneous rank-and-energy matched control.** Per (item, layer, position), a random subspace matching the J arm's achieved effective rank and removed energy exactly, orthogonal to the protected rows [mc-dev-validation-olmo31-think-v2]. At matched dose, direction content is the only difference between arms.
- **One unit system.** Two tokenizer-convention faults (BOS units; concatenation) caught by the preregistered stop rule before any confirmatory outcome was viewed [AMENDMENT_1_BOS_UNITS].
- **Audited families.** Hand-audited item families replace a string-split accident; every clustered statistic runs on the audited map.
- **Positive controls.** G4 swap controls certify each lens causally (flip rate 0.76–0.78 vs 0.18–0.24 random) [r5-swap-positive-control-*-v2].

III. The Phase 2 Confirmatory Matrix

[TODO: Frozen Phase 2 results, stated with the thin-two-hop limitation first. Primary outcomes: HP1 composed-minus-direct contrast -0.504 $[-0.720, -0.295]$, $p_{\text{holm}} = 5 \times 10^{-4}$; HP3 J-specific tail excess $+0.279$ $[+0.205, +0.361]$ (Qwen) [n6-confirmatory-analysis-v2]. Held-out replication: HP3 replicates ($+0.297$); HP1 inconclusive on the 9-item two-hop leg. Corrected capacity: Qwen at 5–6% centered excess (lower edge of the reported Claude band), the OLMo 3.1 pair flat at $\sim$1.2% [r2-occupancy-*-v2].]

IV. Label Protection Is Not Span Protection

The opening question of Phase 3 — does protecting output token *labels* from selection keep the removed span away from output *geometry*? — has a sharp answer: no, on every model tested. Under label protection the selected J span overlaps the protected span (projector overlap 0.89/0.94/1.37 for Think/Instruct/Qwen), a large share of removed activation energy sits inside the protected span (37%/42%/28%), and the never-selectable answer direction loses 18–26% of its norm [p3-span-audit-cross-model-v1] (PHASE3-DEVELOPMENT). A span-safe arm zeroes the overlap exactly.

The behavioral consequence decomposes differently per model (Fig. 1): on the OLMo pair the label-protected and span-safe arms damage substantially *different* items (per-item $r = 0.20/0.29$) — a reallocation — while on Qwen $r = 0.75$ and the leakage-dose control is nil: Qwen's label-protected effect was mostly content all along.

i. The specificity claim survives the repair

The campaign's central specificity claim was re-established at confirmatory tier with the leak removed by construction: on Qwen, the span-safe J-specific tail-rate excess over the exact matched control is $+0.096$ (188 items, 26 families, $p \approx 10^{-5}$) [p3-locked-analysis-v1] (PHASE3-CONFIRMATORY), and independently replicates at $+0.102$ on disjoint held-out families [p3-replication-analysis-v1] (PHASE3-REPLICATION) — about one third of the label-protected effect, quantitatively consistent with the leakage/content decomposition (Figs. 2, 3).

V. Controlled Composition

[TODO: P3-P1 on the thick paired bank: contrast -0.261 nats, wild-cluster CI $[-0.52, -0.01]$, $p = 0.062$ at 17 intersection families — directional under the disclosed MDE, an honest near-miss carried by the estimation targets [p3-locked-analysis-v1]. Replication side -0.197, $p = 0.218$, same sign. Think-vs-Instruct on the thick bank indistinguishable ($+0.01$ $[-0.45, +0.39]$). The workspace-word test: Qwen's contrast does NOT appear on the in-context bank S ($+0.02$ $[-0.33, +0.42]$) — per the pre-stated

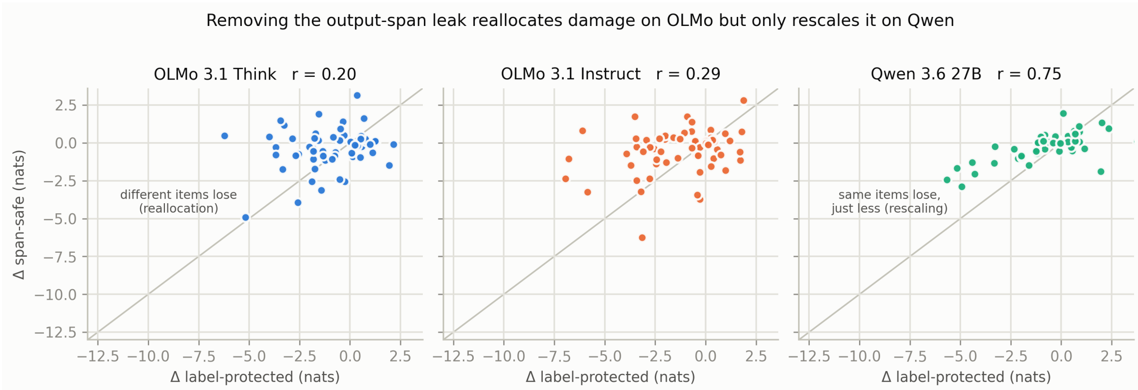

Figure 1: *Per-item damage under the paper-faithful label-protected arm (x) vs the leak-free span-safe arm (y), on the 60 frozen Phase 2 confirmatory items per model. Removing the output-span leak* reallocates *damage on OLMo (r* $= 0.20/0.29$*) and merely* rescales *it on Qwen (r* $= 0.75$*)* `[p3-span-audit-cross-model-v1]` (PHASE3-DEVELOPMENT).

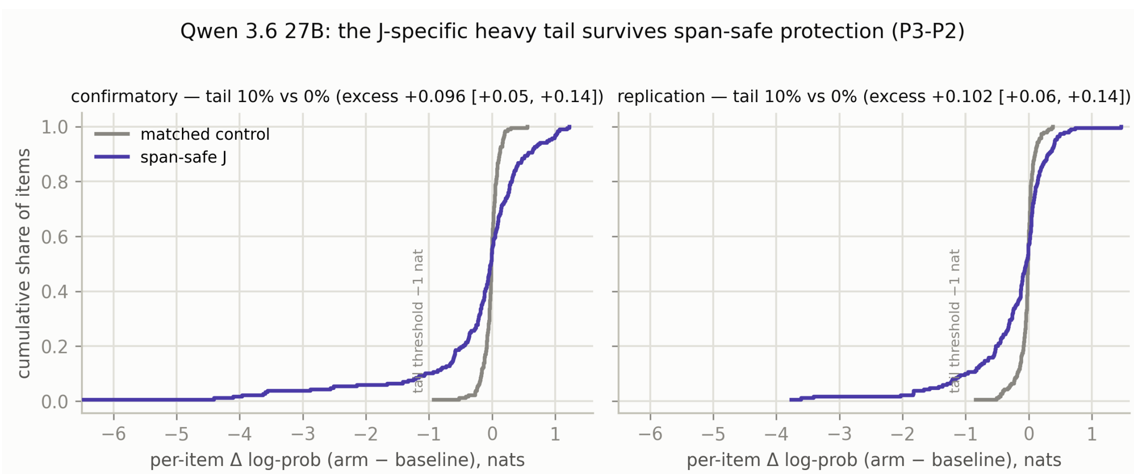

Figure 2: *Per-item damage distributions on Qwen under span-safe J ablation vs the exact rank-and-energy matched control, confirmatory (left) and held-out replication (right). The control produces no* -1*-nat tail; the J arm's tail survives full span protection* `[p3-locked-analysis-v1]`, `[p3-replication-analysis-v1]`.

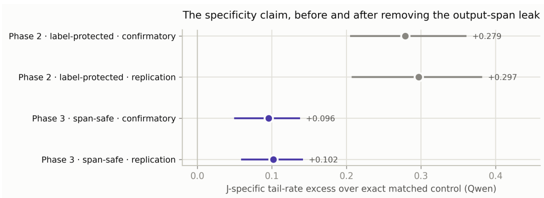

Figure 3: *The specificity estimand across eras: label-protected (Phase 2) vs span-safe (Phase 3), confirmatory and replication sides. The effect shrinks* $\sim 3\times$ *when the output-span leak is removed — and survives with* $p \approx 10^{-5}$*. Phase 3 CIs are family-clustered bootstrap (this paper's re-analysis).*

decision rule the supportable noun is "knowledge-access channel."]

VI. Mechanism: Bridge Mediation

The preregistered mediation primary (P3-P3, model chosen magnitude-blind) rejects on Qwen: protecting the true bridge's token directions rescues composed-item damage by $+0.431$ nats over protecting a matched counterfactual bridge (94 composed items, $p_{\text{holm}} = 0.018$) `[p3-locked-analysis-v1]` (phase3-confirmatory). The seven-arm factorial (Fig. 4) gives the mechanism-level picture:

- **Qwen consumes its bridge channel.** Deleting only the bridge's directions — output fully preserved — damages composed answers (-0.89, CI-clean); substituting the *counterfactual* bridge's direction at matched energy is catastrophically worse than deletion (-4.05): the model reads what sits in the channel and follows it. The unrelated-content lesion is nil (-0.04) `[p3-bridge-mediation-qwen36-27b-v1]` (phase3-development).
- **Think does not.** The swap is inert (-0.09), the rescue contrast mildly negative (-0.20 $[-0.38, -0.04]$) `[p3-bridge-mediation-olmo31-think-v1]`.

i. The accessibility organization

[TODO: The other half of the two-cluster picture: OLMo damage is organized by output accessibility ($\rho = -0.30/-0.31$, CI-clean), Qwen's by composition with the opposite sign ($+0.24$) `[p3-overlap-mining-v1]` (phase3-development). Selection pressure is a $12\times$ model property (blocked rows/position: Think 1.33, Instruct 1.22, Qwen 0.10) yet uncorrelated with damage within model — pressure and span geometry are dissociable. Fig. 5.]

VII. Prose Costs and Selectivity

[TODO: Workstream C: the exact matched control Phase 2 never ran on prose. Prose cost is J-content everywhere, never dose (exact control $+0.002$–$+0.021$/token vs label arm $+0.175$–$+0.945$); the prose-cost ordering inverts the task ordering and tracks projector overlap; span-safe removes 72–78% of the label cost. No model earns "selective" wording at this tier `[p3-prose-grid-figure-v1]` (phase3-development).]

VIII. Reproducibility

[TODO: The N8 ladder as a release gate: Level 1 narrative-blind analysis reproduction ($\leq 5 \times 10^{-4}$ on all four primaries) `[p3-n8-level1-repro-v1]`; Level 2 sentinel cells bit-exact on all three models `[p3-n8-level2-*-v1]`; Level 3 one full 188-item Qwen GPU cell bit-exact `[p3-n8-level3-qwen36-27b-v1]` (methods). Also: single-command repro, append-only evidence registry, frozen partitions, preregistration with disclosed deviations.]

IX. Discussion

[TODO: What is established: a readable, causally load-bearing, content-specific J-channel on Qwen (knowledge-access channel), with bridge mediation; an accessibility-organized content-channel response on the OLMo pair; the instrument lesson (label vs span protection) as a portable contribution

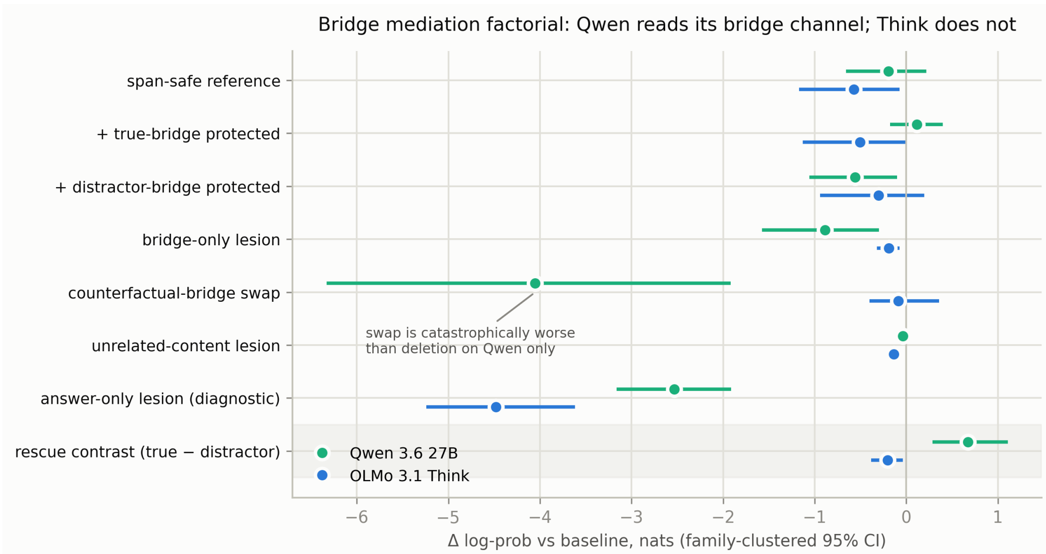

Figure 4: *The bridge-mediation factorial (family-clustered 95% CIs; 40 Qwen / 20 Think composed items, span-safe base protection). The same intervention grammar that shows Qwen routing composed answers through readable bridge content shows Think not doing so* `[p3-bridge-mediation-*-v1]` (PHASE3-DEVELOPMENT).

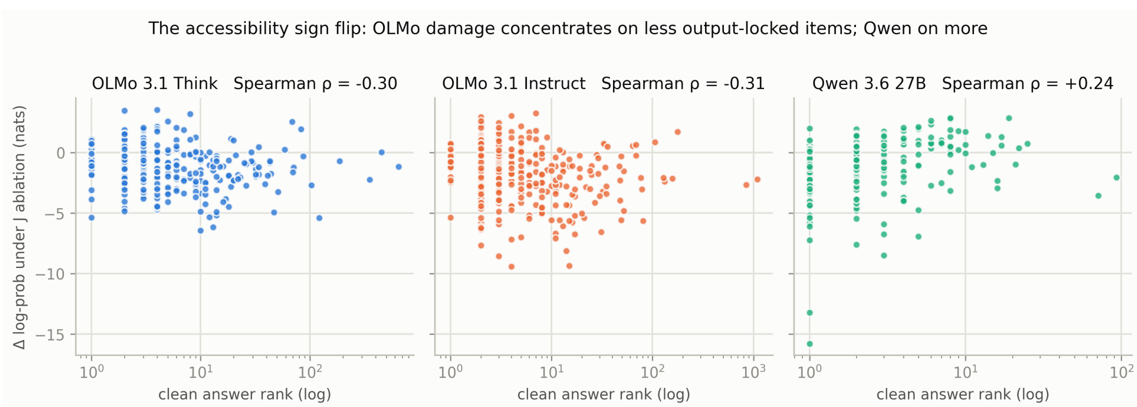

Figure 5: *Accessibility organization of J-ablation damage (975 frozen Phase 2 items). On the OLMo pair, items whose answers are* less *output-locked lose more; on Qwen the sign flips* `[p3-overlap-mining-v1]` (PHASE3-DEVELOPMENT).

for anyone using protected dynamic ablation. What is not: the working-memory reading of "workspace" (fails the bank-S test on both clusters); capacity-causes-shape (three assay-valid models); any claim about consciousness. Limitations: MDE on P3-P1; development-tier arms of the factorial; two model families.]

[TODO: Related work section (or fold into §I).]

References

[Anthropic, 2026] Anthropic Interpretability Team. *Verbalizable Representations Form a Global Workspace in Language Models.* transformer-circuits.pub/2026/workspace, July 2026.

[Anthropic, 2026] Anthropic Interpretability Team. *jacobian-lens* (reference implementation). github.com/anthropics/jacobian-lens, 2026.

[OLMo Team, 2026] OLMo Team, Allen Institute for AI. *OLMo 3.1: Open Language Models.* **[TODO: cite the actual OLMo 3.1 report]**

[Qwen Team, 2026] Qwen Team. *Qwen 3.6 Technical Report.* **[TODO: cite the actual Qwen 3.6 report]**

AI Disclosure

[TODO: Adapt the standard disclosure: AI assistants used for run orchestration, analysis tooling, formatting, and proofreading; all design decisions, experimental results, and final content are my own.]

Code and Data Availability

Analysis code, the evidence registry, and small metrics are in the `labs` repository under `interpretability/` (packages `jspace_part2`, `jspace_phase3`); heavy artifacts (fitted lenses, per-item parquets, generation traces) are archived in the run store. Every primary quantity regenerates via the packages' single-command reproduction entry points.